TELCO CUSTOMER CHURN ANALYSIS

Business Analysis Report

Dataset: 7,032 Validated Customer Records | Tools: Python · Pandas · Matplotlib · Seaborn

Task	Title	Key Focus
Task 1	Understand the Dataset	Data loading, first look, data types, missing values
Task 2	Data Cleaning	Handle nulls, remove duplicates, standardize columns
Task 3	Exploratory Data Analysis (EDA)	Summary stats, distributions, churn rates, associations
Task 4	Customer Segmentation Visualization	Tenure segments, donut chart, clustered bar chart
Task 5	Advanced Analysis	Demographics, contracts, payments, lifecycle analysis

TASK 1 — Understanding the Dataset

Objective: Load the Telco Customer Churn dataset, display the first 10 rows, identify data types for each column, and check for missing values.

1.1 Dataset Overview

Property	Value
Total Records (raw)	7,043 customers
Total Columns	21 features
Target Variable	Churn (Yes / No)
Data Source	Telco Customer Churn Dataset (CSV)
Tool Used	Python — pandas library

1.2 Column Data Types

Column	Data Type	Description
customerID	object (string)	Unique customer identifier
gender	object (string)	Customer gender: Male / Female
SeniorCitizen	int64	Binary: 1 = Senior, 0 = Non-Senior
Partner	object (string)	Has partner: Yes / No
Dependents	object (string)	Has dependents: Yes / No
tenure	int64	Months with the company
PhoneService	object (string)	Has phone service: Yes / No
MultipleLines	object (string)	Multiple phone lines: Yes / No / No phone service
InternetService	object (string)	Internet type: DSL / Fiber optic / No
OnlineSecurity	object (string)	Online security add-on: Yes / No
TechSupport	object (string)	Tech support add-on: Yes / No

Contract	object (string)	Contract type: Month-to-month / One year / Two year
PaymentMethod	object (string)	Payment method (4 options)
MonthlyCharges	float64	Monthly billing amount ($)
TotalCharges	object → float64	Total charges (required conversion)
Churn	object (string)	Target variable: Yes / No

1.3 Missing Values Check

Finding	Detail
Columns with structural nulls	0 columns — no NaN in standard fields
TotalCharges (special case)	11 rows contain whitespace ' ' instead of numeric value
Root Cause	New customers with tenure = 0 had no charges yet
Action Required	Convert TotalCharges to numeric → whitespace becomes NaN → drop 11 rows

Interpretation — Task 1

The dataset is largely clean with only one meaningful data issue: 11 rows where TotalCharges was stored as whitespace rather than a numeric value. These correspond to customers with zero tenure (brand-new accounts). The dataset contains 21 columns covering demographics, service subscriptions, billing, contract details, and the churn outcome — providing a rich foundation for analysis.

TASK 2 — Data Cleaning

Objective: Clean the dataset to ensure it is ready for analysis. Handle missing values, remove duplicates, and standardize column names.

2.1 Cleaning Steps Applied

Step	Action	Code / Detail	Result
1	Convert TotalCharges	pd.to_numeric(df['TotalCharges'], errors='coerce')	11 whitespace values → NaN
2	Drop missing rows	df.dropna(subset=['TotalCharges'])	11 rows removed
3	Duplicate check	df.duplicated().sum()	0 duplicates found
4	Standardize columns	str.lower().str.replace(' ','_')	All columns lowercase with underscores
5	Create binary target	churn_binary: Yes=1, No=0	Used for numerical correlation analysis

2.2 Dataset Before vs After Cleaning

Metric	Before Cleaning	After Cleaning
Total Rows	7,043	7,032
Rows Removed	—	11 (0.16%)
Missing Values (TotalCharges)	11	0
Duplicate Records	0	0
Column Name Format	Mixed case with spaces	Lowercase with underscores
TotalCharges Data Type	object (string)	float64 (numeric)

Records After Cleaning	Rows Dropped	Duplicates Found
7,032	**11**	**0**
Valid for analysis	Only 0.16% of data	Dataset is unique

Interpretation — Task 2

The cleaning process was minimal and did not meaningfully affect the dataset. Removing 11 rows (0.16%) introduces no material bias. The decision to drop rather than fill these rows is justified because TotalCharges for zero-tenure customers is genuinely missing — there is no reasonable value to impute. After cleaning, the dataset is fully numeric where required, free of duplicates, and consistently formatted for analysis.

TASK 3 — Exploratory Data Analysis (EDA)

Objective: Understand the distribution and patterns in the cleaned data through summary statistics, visualizations, and churn rate analysis across all key variables.

3.1 Summary Statistics — Numerical Columns

Column	Mean	Median	Mode	Std Dev	Min	Max	Skewness
Tenure	32.42	29.00	1.00	24.55	1	72	0.24 (near-normal)
MonthlyCharges	64.80	70.35	20.05	30.09	18.25	118.75	-0.22 (slight left)
TotalCharges	2,283.30	1,397.48	20.20	2,266.77	18.80	8,684.80	0.96 (right-skewed)

3.2 Chart 1 — Overall Churn Distribution

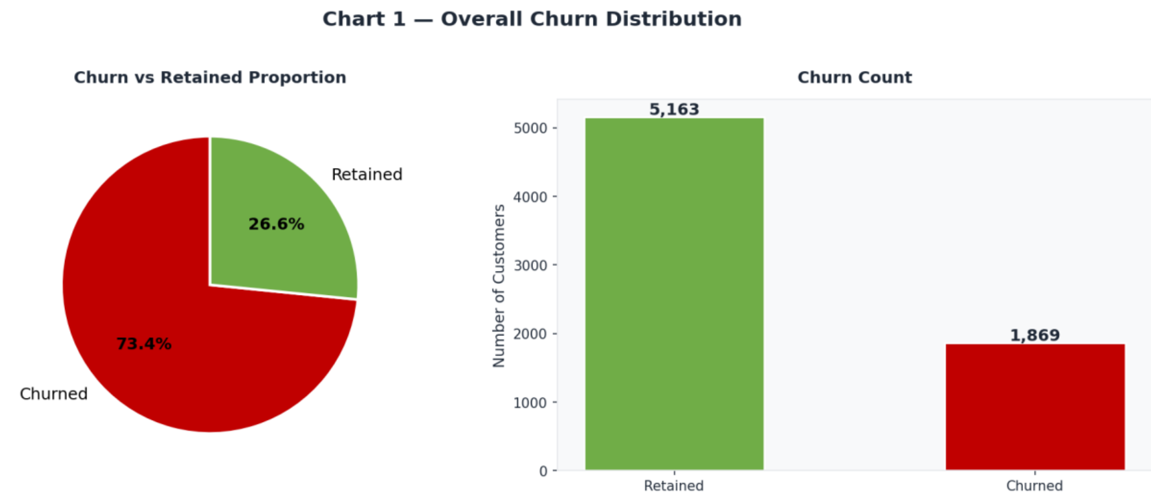

Chart 1: Left — Churn proportion pie chart. Right — Absolute count of retained vs churned customers.

Total Customers	Retained (No Churn)	Churned
7,032	**5,163**	**1,869**
After cleaning	73.42%	26.58%

Interpretation — Chart 1

The overall churn rate is 26.58%, meaning approximately 1 in 4 customers left the company. While the majority are retained, losing more than a quarter of the customer base represents a significant business problem. The dataset is moderately imbalanced (73:27) but the churn group is large enough for robust analysis. This churn level justifies deeper investigation into which customer segments are most at risk.

3.3 Chart 2 — Numerical Variable Distributions

Chart 2: Histograms (top row) and box plots (bottom row) for tenure, MonthlyCharges, and TotalCharges.

Interpretation — Chart 2

Tenure shows a bimodal distribution — many customers with very short tenure (1-2 months) and many with long tenure (60-72 months), suggesting two customer types: new undecided customers and committed long-term ones. MonthlyCharges is broadly distributed with peaks at low (~$20) and high (~$80-90) values, reflecting distinct pricing tiers. TotalCharges is right-skewed because it accumulates over time — long-tenure, high-paying customers naturally generate the highest totals.

3.4 Chart 3 — Numerical Variables by Churn Status

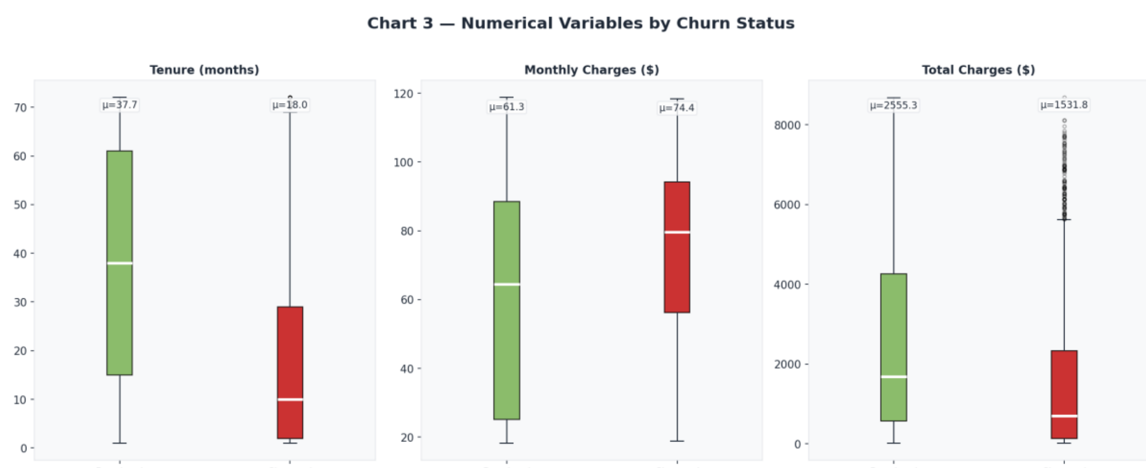

Chart 3: Box plots comparing tenure, MonthlyCharges, and TotalCharges split by churn outcome.

Metric	Retained (No Churn)	Churned (Yes)	Difference
Avg Tenure	37.65 months	17.98 months	Churned customers stay 2x less
Median Tenure	38 months	10 months	Stark gap — churned customers leave early
Avg Monthly Charge	$61.31	$74.44	Churned customers pay 21% more
Avg Total Charges	$2,555.34	$1,531.80	Lower because they leave sooner

Interpretation — Chart 3

Three critical findings emerge: (1) Churned customers have much shorter tenure — median of only 10 months vs 38 months for retained customers. This is the strongest individual signal. (2) Churned customers actually pay MORE per month on average ($74.44 vs $61.31). This suggests they are not leaving because they are on cheap plans — they are leaving despite paying a premium, implying perceived value is the issue. (3) Lower total charges for churned customers is a consequence of early departure, not a cause.

3.5 Chart 4 — Churn by Contract, Internet & Payment Method

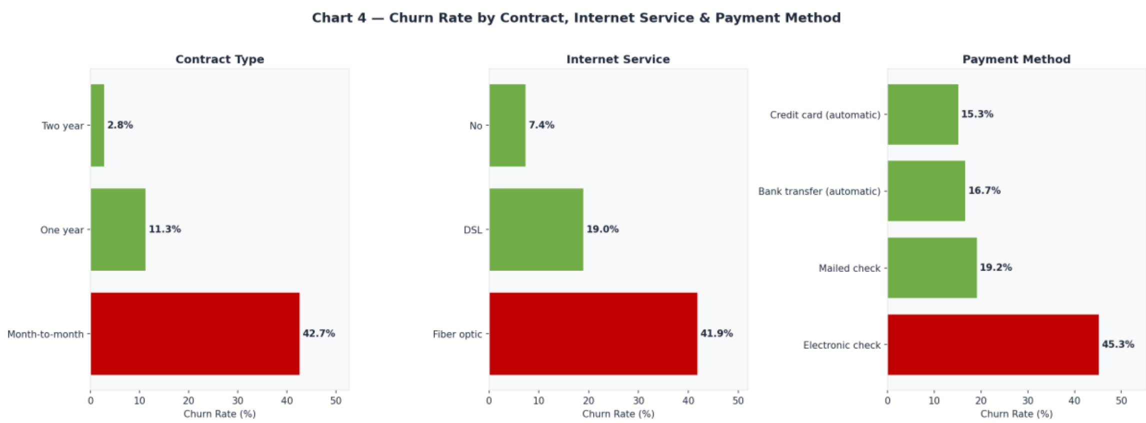

Chart 4: Horizontal bar charts showing churn rates by contract type, internet service, and payment method.

Category	Group	Churn Rate	Key Finding
Contract Type	Month-to-month	42.71%	Highest — no commitment required
Contract Type	One year	11.28%	74% lower than month-to-month
Contract Type	Two year	2.85%	Almost zero churn
Internet Service	Fiber optic	41.89%	High value, high risk
Internet Service	DSL	19.00%	Moderate — more stable
Internet Service	No internet	7.43%	Lowest churn
Payment Method	Electronic check	45.29%	Highest — weak commitment signal

Payment Method	Mailed check	19.20%	Moderate
Payment Method	Bank transfer (auto)	16.73%	Low — auto-pay = stable
Payment Method	Credit card (auto)	15.25%	Lowest — most stable

Interpretation — Chart 4

Contract type is the single strongest churn driver in the dataset. Month-to-month customers churn at 15 times the rate of two-year contract holders (42.71% vs 2.85%). Fiber optic customers have unexpectedly high churn despite similar tenure to DSL customers, suggesting a service quality or value perception problem. Electronic check users churn at 45.29% — nearly 3 times the rate of automatic payment users, likely reflecting financial instability or lower engagement.

3.6 Chart 5 — Add-On Services & Demographics

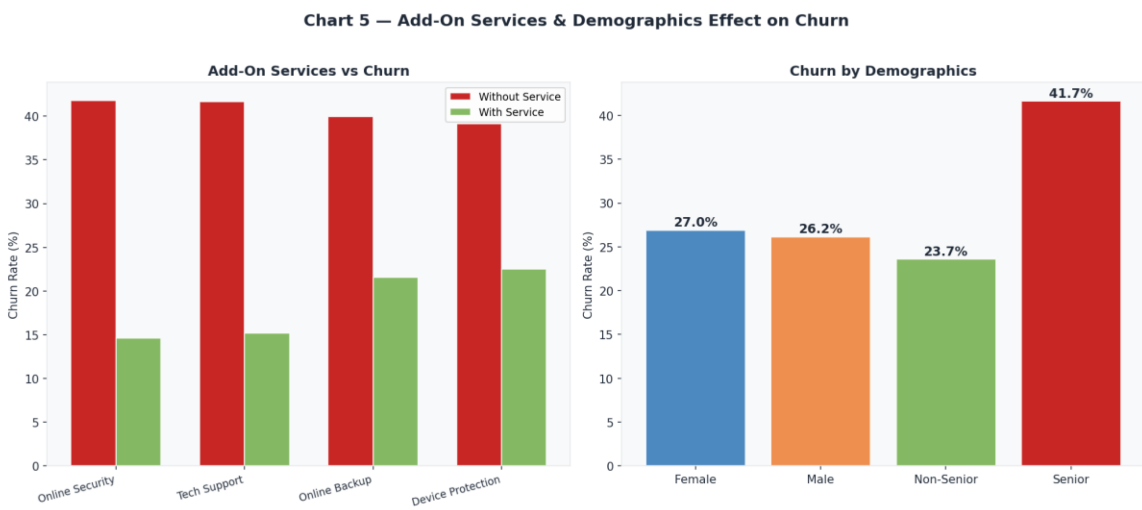

Chart 5: Left — Churn rates for customers with/without add-on services. Right — Churn by gender and senior status.

Feature	Cramer's V	Highest-Risk Group	Churn Rate
Contract	0.4096	Month-to-month	42.71%
OnlineSecurity	0.3470	No security	41.78%
TechSupport	0.3425	No tech support	41.65%
InternetService	0.3219	Fiber optic	41.89%
PaymentMethod	0.3030	Electronic check	45.29%
OnlineBackup	0.2919	No backup	~39.9%
DeviceProtection	0.2816	No device protection	~39.1%

Interpretation — Chart 5

Customers without online security and tech support churn at over 41% — nearly double the overall rate. This reveals that churn is driven not just by contract type and billing, but also by perceived service safety and support. Customers who subscribe to protection and support services feel more confident in their provider. Senior citizens churn at 41.7% vs 23.6% for non-seniors, a 76% relative difference that warrants a dedicated retention strategy.

3.7 Chart 6 — Correlation Heatmap

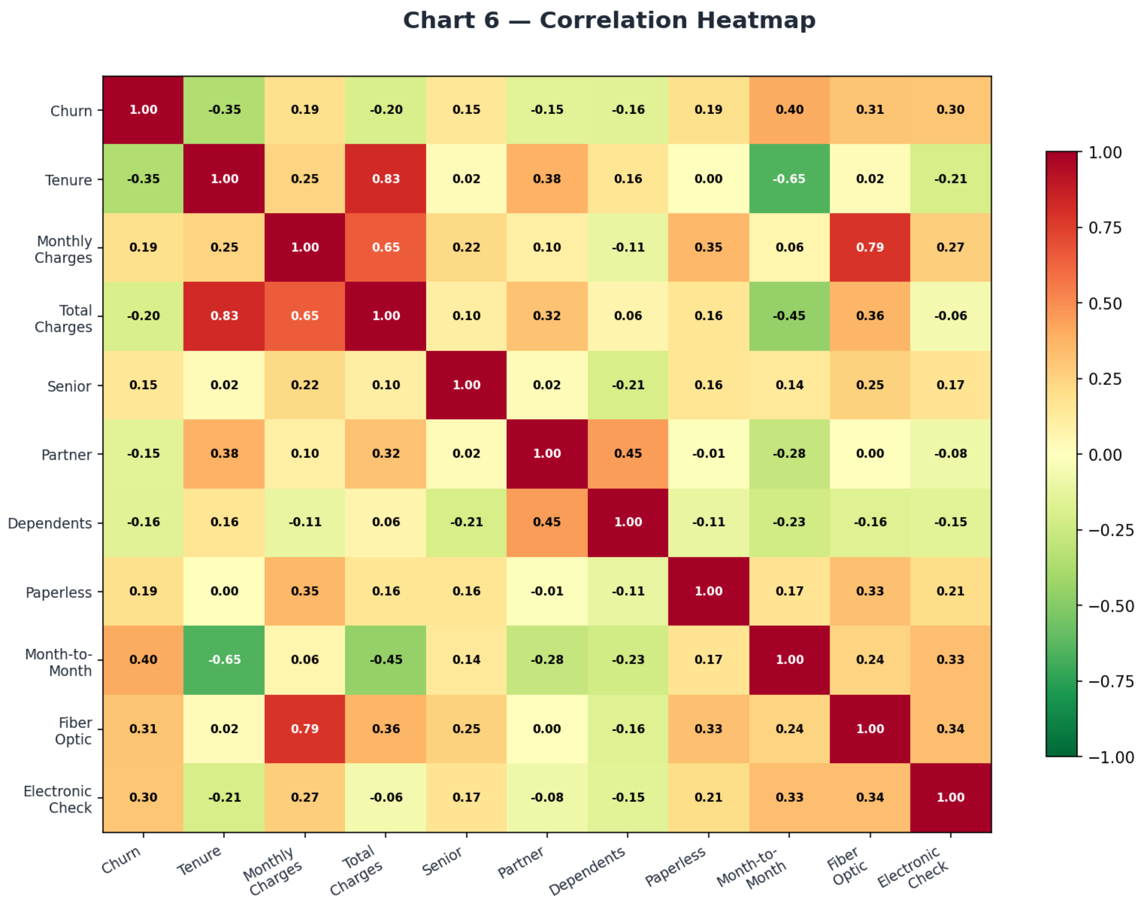

Chart 6: Correlation heatmap showing relationships between numerical and binary-encoded variables.

Feature	Correlation with Churn	Direction	Interpretation
tenure	-0.354	Negative	Longer tenure = much less likely to churn
Month-to-Month (binary)	~+0.40	Positive	Strongest positive driver
Electronic Check (binary)	~+0.30	Positive	High churn risk signal
MonthlyCharges	+0.193	Positive	Higher charges = slightly higher churn risk
TotalCharges	-0.199	Negative	High total = long-tenure customer, lower risk
Fiber Optic (binary)	~+0.31	Positive	Fiber customers churn more
Partner (binary)	-0.15	Negative	Having a partner = lower churn tendency

Interpretation — Chart 6

The heatmap confirms that tenure has the strongest numerical correlation with churn (-0.354). The positive correlations of Month-to-Month, Fiber Optic, and Electronic Check with churn align with the bar chart findings. Notably, TotalCharges shows a negative correlation not because high charges reduce churn, but because long-tenure customers accumulate high totals — it is a proxy for tenure. The partner variable being negatively correlated with churn suggests that household-level stability reduces the likelihood of switching providers.

3.8 Chart 7 — Combined High-Risk Segment

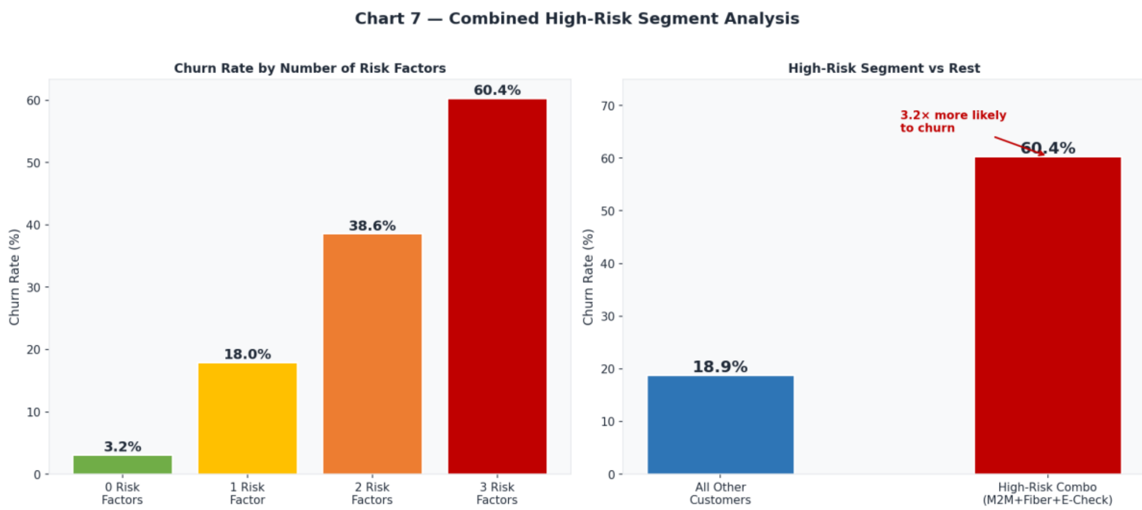

Chart 7: Left — Churn rate by number of stacked risk factors. Right — High-risk combo segment vs all others.

Risk Factors	Customers	Churned	Churn Rate	Avg Monthly	Avg Tenure
0 factors	1,962	62	3.16%	$44.72	46.74 mo
1 factor	2,111	380	18.00%	$61.04	31.53 mo
2 factors	1,652	638	38.62%	$75.73	26.89 mo
3 factors (all)	1,307	789	60.37%	$87.18	19.37 mo

Most Important Finding — Combined High-Risk Segment

Customers with ALL THREE risk factors (Month-to-month contract + Fiber optic internet + Electronic check payment) churn at 60.37% — compared to just 18.86% for all other customers. This group of 1,307 customers represents the single highest-priority retention target. They are new (avg 19 months tenure), pay the most ($87.18/month), and are leaving at a critical rate. A targeted retention campaign combining a contract upgrade offer, auto-pay incentive, and service bundle could dramatically reduce churn in this group.

TASK 4 — Customer Segmentation Visualization

Objective: Segment customers by tenure into three groups (New: 0-12 months, Mid: 13-36 months, Loyal: 37+ months), visualize their distribution, and compare average monthly charges and churn rates across segments.

4.1 Chart 8 — Tenure Segmentation (Donut + Clustered Bar)

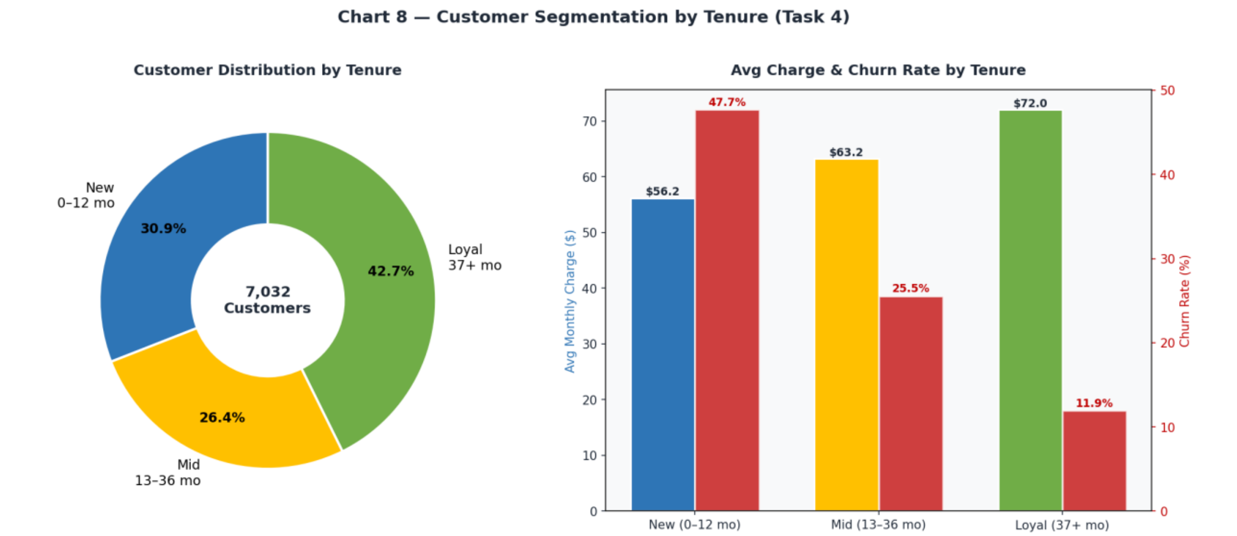

Chart 8: Left — Donut chart showing customer distribution by tenure segment. Right — Clustered bar comparing avg monthly charge vs churn rate.

New (0–12 Mo)	Mid (13–36 Mo)	Loyal (37+ Mo)
2,186 | 31.0%	**1,856 | 26.4%**	**3,001 | 42.6%**
Churn Rate: 47.4%	Churn Rate: 25.5%	Churn Rate: 11.9%

4.2 Segment Summary Table

Segment	Customers	Share	Avg Monthly	Avg Total Charges	Churn Rate	Risk Level
0–12 Mo (New)	2,186	31.0%	$56.10	$276.62	47.4%	HIGH
13–36 Mo (Mid)	1,856	26.4%	$63.25	$1,513.54	25.5%	MEDIUM
37+ Mo (Loyal)	3,001	42.6%	$72.01	$4,213.72	11.9%	LOW

Interpretation — New Customers (0–12 months): CRITICAL

Nearly 1 in 2 new customers churns within the first year (47.4%). This is the most dangerous segment. These customers are paying the least ($56.10/month) but leaving at the highest rate, suggesting they have not yet found enough value in the service to stay. Early onboarding, first-year engagement programs, and loyalty incentives are critical here.

Interpretation — Mid Customers (13–36 months): WATCH ZONE

Mid-tier customers show meaningful improvement (25.5% churn) but are still above the overall average. They are paying more ($63.25/month) and have invested some time in the relationship. This is the decisive window — targeted contract upgrade offers and milestone rewards at 12, 24, and 36 months can successfully pull this group into the loyal segment.

Interpretation — Loyal Customers (37+ months): PROTECT

Loyal customers are the company's revenue foundation. They churn at only 11.9%, pay the most ($72.01/month), and generate an average of $4,213 in total lifetime charges — over 15 times the average new customer ($276). Every retained new customer has the potential to become a loyal customer worth 15x more. Protecting this segment through VIP programs and proactive outreach is essential.

TASK 5 — Advanced Analysis

Objective: Perform multi-dimensional churn analysis across demographics (gender, senior status), behavioral factors (contract type, payment method), and lifecycle stages. Identify patterns and derive targeted business insights.

5.1 Chart 9 — Advanced Demographic & Behavioral Analysis

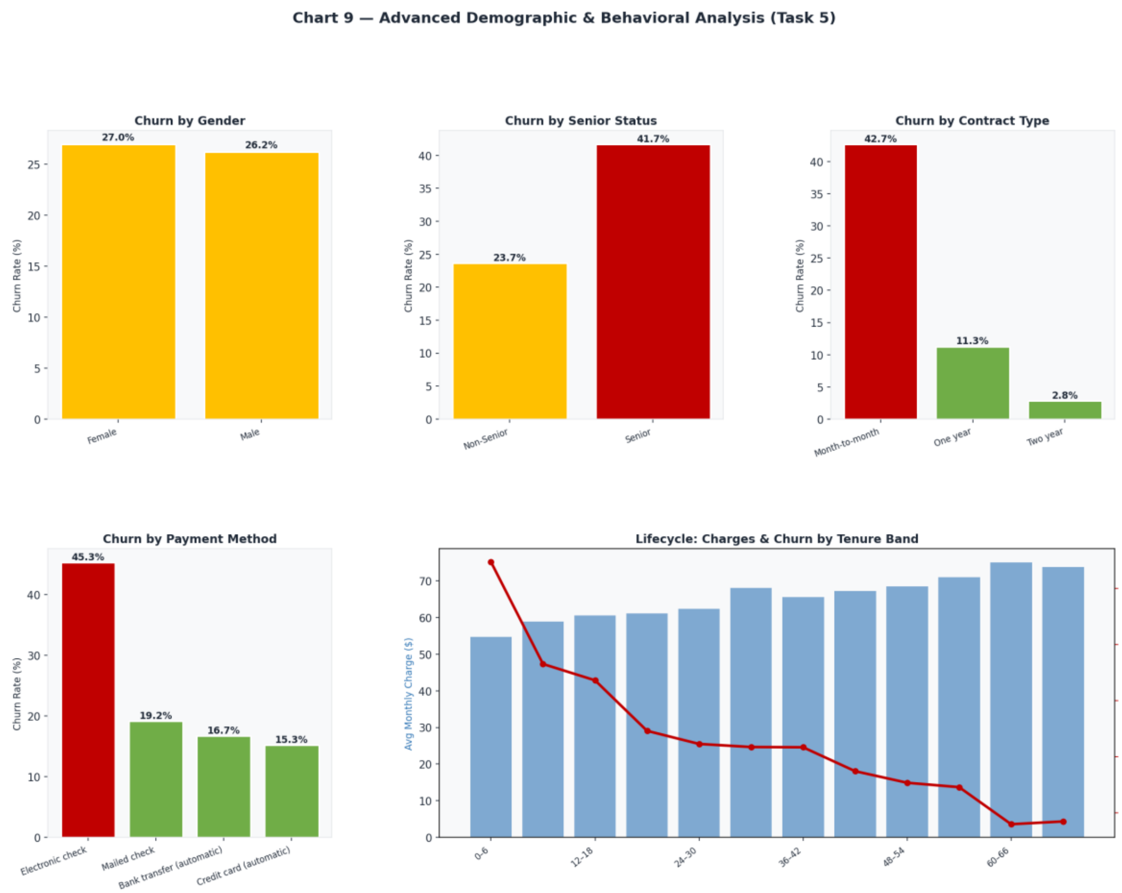

Chart 9: Six-panel analysis covering gender, senior status, contract type, payment method, and lifecycle trend.

5.2 Churn by Gender

Female Churn Rate	Male Churn Rate	Difference
26.9%	**26.2%**	**0.7%**
3,488 customers	3,544 customers	Statistically negligible

Interpretation — Gender

Gender has virtually no impact on churn (26.9% vs 26.2%, a difference of 0.7%). This is one of the most important negative findings: gender-based targeting for retention campaigns would be a waste of resources. The company should focus its retention budget on contract type, payment method, and tenure — all of which have far greater predictive power.

5.3 Churn by Senior Citizen Status

Non-Senior Churn	Senior Citizen Churn	Relative Risk
23.6%	**41.7%**	**+ 76%**
5,901 customers	1,131 customers	Seniors churn far more

Interpretation — Senior Citizens

Senior citizens churn at 41.7% — 76% higher than non-seniors (23.6%). This is a significant and actionable demographic finding. Possible drivers include: difficulty navigating digital interfaces, complex billing, lack of tailored support, or fixed-income sensitivity to monthly charges. Recommended actions: simplified billing statements, a dedicated senior support line, technology assistance programs, and senior-specific contract packages at competitive rates.

5.4 Churn by Contract Type

Contract Type	Customers	Churned	Churn Rate	Avg Tenure	Risk vs 2-Year
Month-to-month	3,875	1,655	42.71%	18.04 months	15x higher
One year	1,472	166	11.28%	42.07 months	4x higher
Two year	1,685	48	2.85%	57.07 months	Baseline

Interpretation — Contract Type (Most Impactful Finding)

Contract type is the single most powerful churn predictor in the entire dataset (Cramer's V = 0.41). Month-to-month customers can leave at any time with no financial penalty — and they do, at a rate of 42.71%. Two-year contract holders churn at just 2.85%. The business implication is clear: every month-to-month customer converted to a one-year or two-year contract dramatically reduces their churn probability. Even partial success in this conversion strategy would deliver measurable revenue retention.

5.5 Churn by Payment Method

Payment Method	Customers	Churn Rate	Type	Risk Signal
Electronic check	2,365	45.29%	Manual	Highest risk — 3x auto-pay
Mailed check	1,604	19.20%	Manual	Elevated — manual friction
Bank transfer	1,542	16.73%	Automatic	Low — committed

(auto)				and stable
Credit card (auto)	1,521	15.25%	Automatic	Lowest — most engaged

Interpretation — Payment Method

Automatic payment users churn at roughly 16% while electronic check users churn at 45.29% — nearly three times higher. This is not just a payment preference: it is a commitment signal. Customers who set up automatic payments have made a friction-bearing decision to stay. Electronic check users maintain maximum flexibility to leave. Offering a billing credit ($5-10/month) or a one-time incentive to customers who switch to automatic payment is one of the quickest and most cost-effective retention levers available.

5.6 Lifecycle Analysis — Charges & Churn Across Tenure Bands

Interpretation — Lifecycle Pattern (Chart 9, bottom panel)

Two clear and opposite trends run simultaneously across the customer lifecycle: (1) Average monthly charges rise steadily from ~$50 at 0-6 months to ~$73 at 66-72 months — demonstrating that the company successfully upsells services as customers mature. (2) Churn rate falls sharply from approximately 55-60% in the first 6-month band to under 5% after 42 months. The first 12 months represent a critical danger zone where churn is catastrophically high despite lower charges. Once a customer survives beyond 18-24 months, the probability of long-term retention increases dramatically. This lifecycle pattern has a direct implication: every dollar invested in early retention yields compounding returns as that customer moves into higher-paying, lower-churn tenure bands.

FINAL SUMMARY & BUSINESS RECOMMENDATIONS

Summary of Key Findings

#	Finding	Key Metric	Priority
1	New customers (0–12 months) churn at nearly 50%	47.4% churn rate	CRITICAL
2	Month-to-month contract is the top churn predictor	42.71% vs 2.85%	CRITICAL
3	Electronic check payment = highest churn risk	45.29% churn rate	CRITICAL
4	Combined high-risk group: M2M + Fiber + E-check	60.37% churn rate	CRITICAL
5	Senior citizens churn 76% more than non-seniors	41.7% vs 23.6%	HIGH
6	Customers without	~41% vs ~15%	HIGH

	security/support churn more		
7	Loyal customers generate 15x more lifetime revenue	$4,213 vs $276	OPPORTUNITY
8	Two-year contracts almost eliminate churn	Only 2.85%	LEVERAGE
9	Auto-payment users churn 3x less than e-check users	16% vs 45%	LEVERAGE
10	Gender has negligible impact on churn	0.7% difference	LOW PRIORITY

Business Recommendations

01. Prioritize the High-Risk Combo Segment (1,307 customers)

Target the Month-to-month + Fiber optic + Electronic check group immediately. With 60.37% churn and $87.18 average monthly charge, retaining even 20% of these customers would save significant annual revenue. Offer: switch to a 1-year contract + enable auto-pay = receive a free security bundle or $20 bill credit. This single campaign hits all three major churn levers simultaneously.

02. Invest in Early Onboarding (0–12 months)

The first 12 months drive nearly half of all churn (47.4%). Introduce a structured onboarding program: welcome calls at Day 7 and Day 30, tutorial resources, a first-year loyalty discount, and proactive check-ins at months 3, 6, and 9. Goal: reduce new-customer churn from 47.4% to below 30%.

03. Convert Month-to-Month Customers to Longer Contracts

Offer month-to-month customers tiered incentives: 10% discount for switching to a 1-year contract, 15-20% for 2-year. Include free add-ons (tech support, online security) as part of the upgrade package. Even converting 30% of the 3,875 month-to-month customers would retain hundreds of additional customers annually.

04. Drive Auto-Payment Adoption

Create a simple, incentivized migration path from electronic check to automatic payment. Options: one-time $10 bill credit, ongoing $3-5/month discount, or simplified in-app auto-pay setup. Electronic check users who switch to auto-pay reduce their churn probability by approximately 30 percentage points.

05. Senior Citizen Retention Program

Design dedicated support for senior customers (41.7% churn): simplified paper bills, a toll-free senior support line, home visit options for tech setup, and senior-specific pricing plans. Reducing senior churn toward the non-senior rate (23.6%) would retain over 200 additional senior customers annually.

06. Bundle Security & Support for Fiber Customers

Fiber optic customers churn at 41.89% despite paying the most ($91.50/month). Investigate service quality complaints and bundle online security + tech support into fiber packages. A 'Fiber Premier' bundle that adds security, backup, and support at a small premium could dramatically improve perceived value and reduce churn.

07. Reward and Protect Loyal Customers (37+ months)

Loyal customers generate $4,213 in lifetime revenue on average. Introduce a VIP loyalty tier at the 36-month milestone: exclusive pricing, dedicated account management, priority support, and anniversary rewards. Preventing even a 5% increase in churn among this segment protects millions in recurring revenue.

SaiKet Systems · Business Analysis Internship · Telco Customer Churn Dataset (n=7,032)

Tools: Python · Pandas · Matplotlib · Seaborn | Analysis by Sandra Abu Abbas